DO NOW — RECALL

Signed binary and shifts

Name: _____ Class: _____ Date: _____

1. Does changing a by-value int parameter change the caller's int?

2. What is a return value?

3. Convert 00001010 from unsigned binary to decimal.

4. Calculate 00001111 + 00000001 using 8 bits.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

No. It is a copy.

2. Model answer

A result a function passes to its caller.

3. Model answer

10.

4. Model answer

00010000, decimal 16.

Vocabulary — know these terms

two's complement — Fixed-width signed binary representation.

sign bit — Leftmost bit indicating sign in two's complement.

logical shift — Shift with zero bits entering vacated positions.

arithmetic right shift — Shift right while extending the sign bit.

overflow — Result cannot be represented in the available bit width.

NEW LEARNING

Read, annotate and explain

POSITIVE 13

+13 written with 8 bits is 00001101.

INVERT BITS

Invert each bit to make 11110010.

ADD ONE

Add one to get 11110011, the 8-bit two's-complement representation of −13.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find −9 in 8-bit two's complement.

Predict first

Write +9 in 8 bits and invert all the bits.

Model the result

Add 1 to the inverted bits: 11110111 represents −9.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write −5 as 8-bit two's complement.

Question 2

Unsigned 8-bit 00110100 is shifted logically right one place. Give the result and decimal value.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

11110100 represents −12 in two's complement. Arithmetically shift right one place and explain the new left bit.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Arithmetic right shift of a negative two's-complement value inserts zero at the left.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.